

Road Safety Infrastructure Facility Standards - Indian Context

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Contents

- AH & NH network in India
- Key Infrastructure Issues affecting Road Safety
- Guidelines – What is available & missing
- Government Initiatives

Asian Highways - Summary

- Agreement by 32 countries
- Total 140,000 km
- In India – 11,525 km of National Highways (NHs) & 26 km of State Roads



Indian Road Network

- Second Largest in the World
- About 65% of the freight and 80% of the passenger traffic are carried by the roads
- National Highways are the highest category roads; constitute only 1.5% of the total road network, but carries 40% of the total traffic
- Notorious for road accidents - > 145,000 fatalities during the year 2015

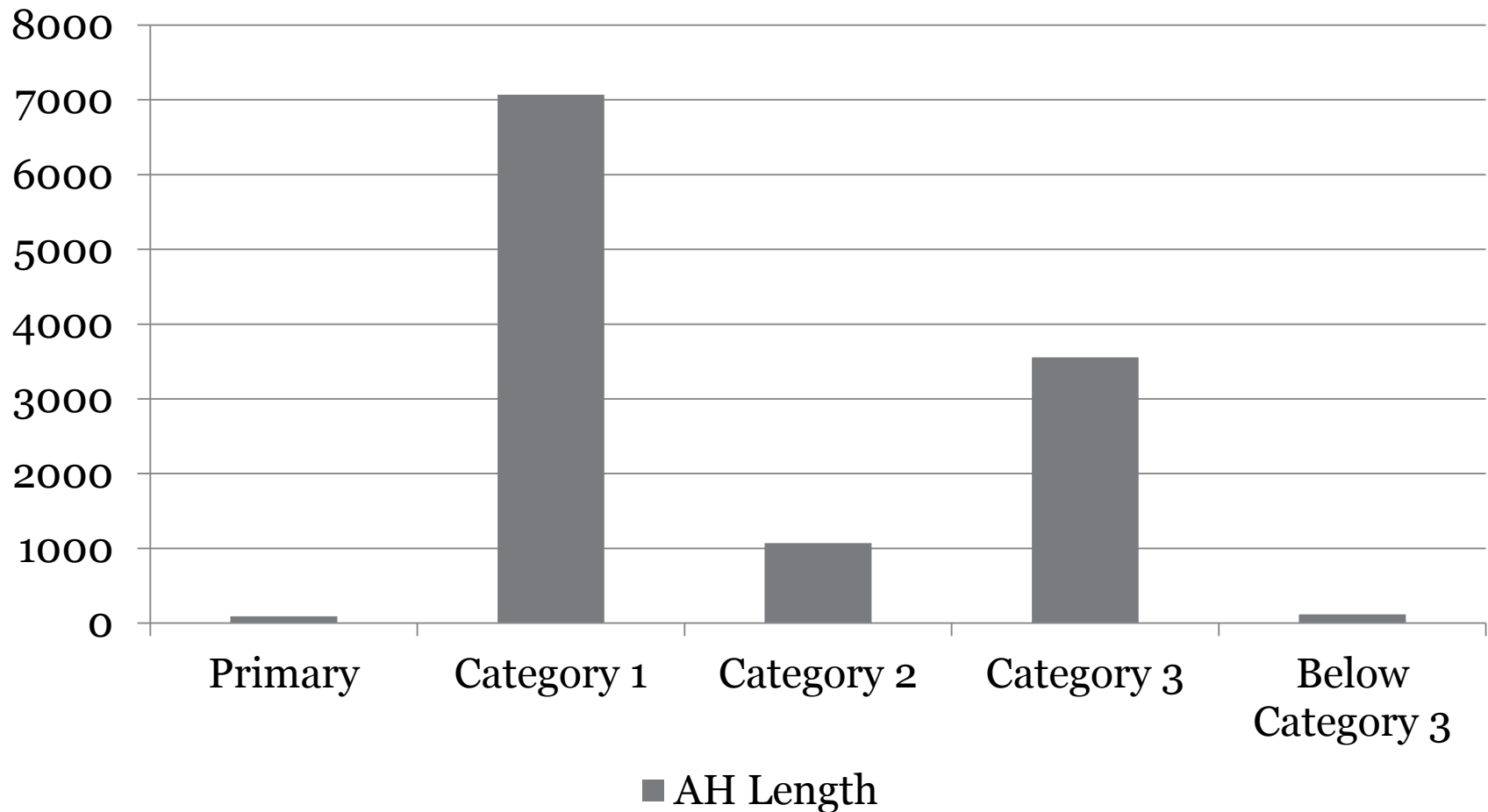
Road Network

Sl No.	Road Type	Managing Agency
1.	Expressways	NHAI/ MORT&H
2.	National Highways	
3.	State Highways	State Road Agencies – <i>Public Works Department/ Roads & Bridges Department/ Roads Authorities/ Corporations</i>
4.	Major District Roads	
5.	Other District Roads	
7.	Village Roads	
8.	Forest Roads	

Highways - Standard Configuration

Sl No .	Road Type	AH Classification
1.	Expressways (access controlled)	Primary
2.	Six Lane Four lane (dual carriageway)	Class I
3.	Two Lane with paved shoulders	Class II/ II
4.	Two Lane	Class II/ III

AH Network in India - Status







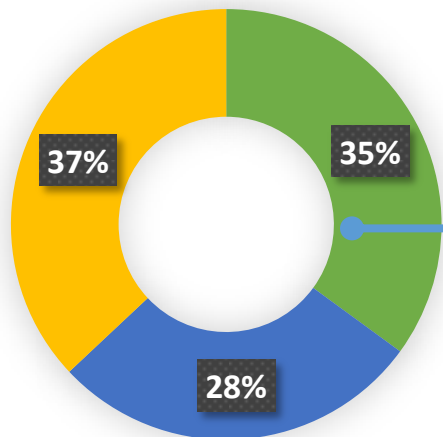
KARUR BYPASS ON NH-7
Aerial View of Four Lane ROB





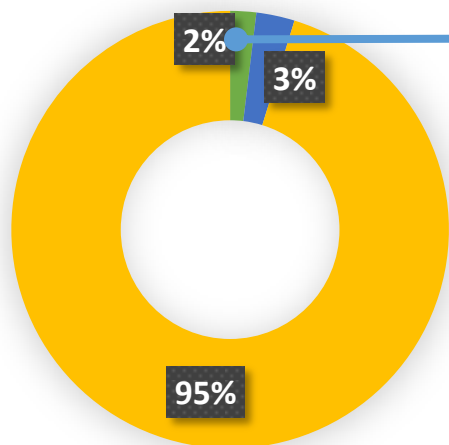


Road Deaths



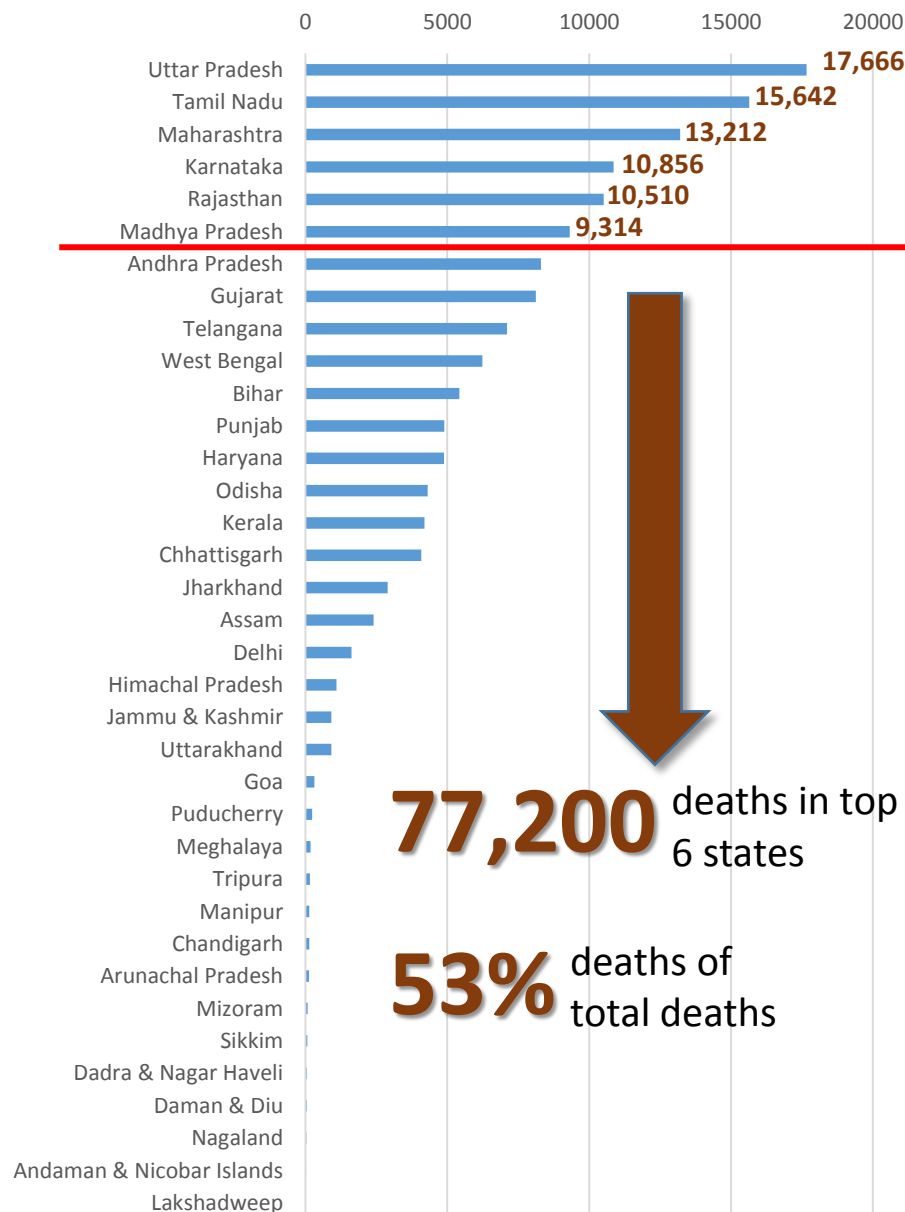
Over 60% of deaths on 5% road network

Road Length



■ National Highways
 ■ State Highways
 ■ Other Roads

Road Deaths in 2015



Safety Engineering - Major Issues

- Land Use/ Uncontrolled Access to the highway
- Median Openings
- Lack of Visibility
- Lack of Footpaths/ Walkways/ Pedestrian Crossing facilities
- Lack of Cycle lanes
- Hazardous Junctions
- Encroachments

Safety Engineering - Major Issues

- Unprotected hazardous objects near the highway
- Improper Speed Management
- Deficiencies in Geometric Design – Sharp Curves, Broken Back Curves, etc.
- Improper/ Lack of Traffic Signs & Road Markings

Geometry for a high speed road



Geometry for a high speed road



08.04.2006

Level difference between the two carriageways



Deficient design and Implementation



Wrong Median Opening leading to Contra-flow



Y – Junctions [Common Problem]

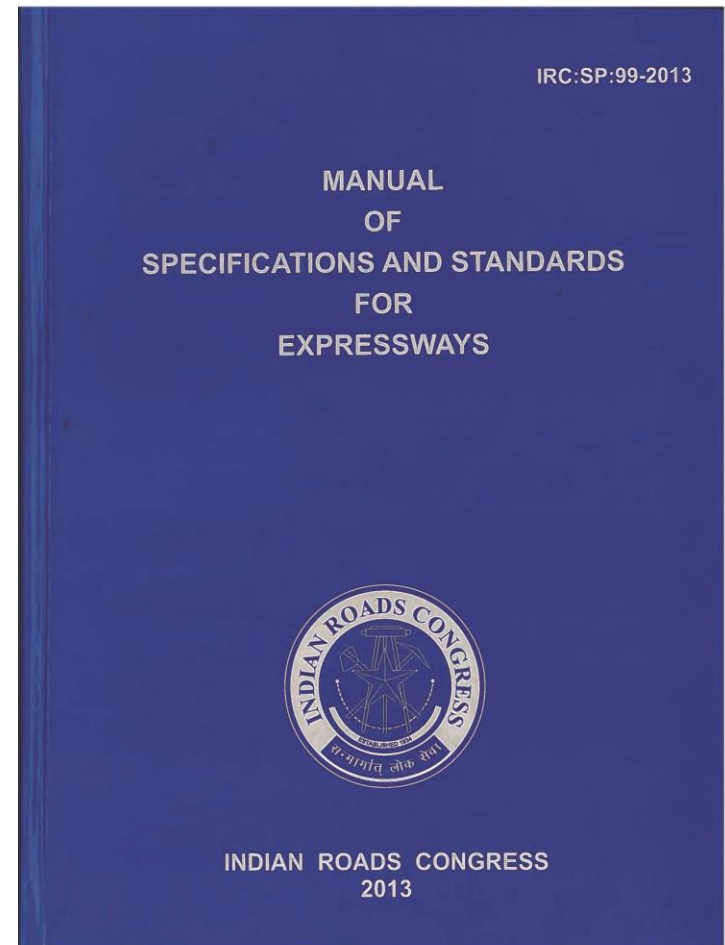


Vulnerable Road Users at Risk



Road Design Guidelines

- Indian Roads Congress (IRC) guidelines used for the planning, design and construction of highways;
- 98 IRC guidelines & 96 special publications covering all aspects of road design and specifications



Road Safety Infrastructure Elements

Guidelines are available for:

- Traffic Signs & Road Markings [Complete]
- Roadside & Median Barriers [needs update]
- Clear zones
- Pedestrian facilities (Crossings, Sidewalks, Refuge Islands & fences)
- Bicycle lanes [focused on Urban]
- Lighting & Sight distance
- Speed hump

Required Guidelines

- Traffic Calming Measures
- Delineators
- At-Grade Intersections
- Roundabouts (in process)
- Bicycle lanes (specific to highways)
- Pedestrian crossings [Not complete]
- Crash Barriers [update required]
- Gateway Designs

Required Infrastructure elements for Safe Travel

- Traffic Calming Measures
- Traffic Signs & Road Markings
[Important]
- Safety barriers at appropriate locations
- Control of Hoardings along the highway

Required Infrastructure elements for Safe Travel

- Controlled Access to the major highway;
- Segregated space for pedestrians & cyclists
- Properly designed intersections
- Strict control of highway encroachments
- Street Lighting on all junctions and other necessary points along the highway

Why we still don't build safe roads?

10. Road safety

While developing the Asian Highway network, Parties shall give full consideration to issues of road safety.

- Agreement has to include specific road safety features to be considered on the network
- Road Safety Audit has to be made mandatory

Government of India Initiatives?

- Large scale blackspot improvements in progress (300 million USD Budget)
- Nation wide training in Road Safety Engineering initiated
- Road Safety Audit is being made mandatory
- Separate funding made available for road safety improvements
- Removal of Hoardings from NH RoW and 10m beyond either side is initiated/ in progress



Thank you.....